

Genesis_Medicine -

16_r15_chromanol_safety_triage

R15 Chromanol Safety-First Fragment Triage: 14-Target Cofolding, ADMET/xTB Filtering, and 30 ns MD Separates Systemic-Safety and Topical-Lead Paths

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Abstract

R15 generated a compact chromanol fragment, OCC1C0c2cc(0)ccc2C1, as a safety-first derivative of the broader pterocarpan-vinyl-polyphenol scaffold program. The central question is whether this fragment should be treated as a topical lead or as a systemic-safety fragment hypothesis. The answer is deliberately split. In ADMET/xTB triage, the R15 parent is predicted to be clean for AMES, DILI, and hERG liability, but its logP is 0.94, below the intended skin-window threshold, and it ranks 11 by the topical composite score. Conversely, the top-3 topical-score analogs satisfy the skin-window heuristic but retain hERG caution values around 0.58-0.70. Boltz-2 14-target cofolding identifies TGFB1, TYR, and DCT as the top three targets by affinity probability, and all three remain stable in 30 ns OpenMM MD with mean ligand RMSD values of 0.51, 0.92, and 0.47 Å. Raw-pose sanity gating found 9 pass, 5 review, and 0 fail rows across the R15 14-target panel. We therefore frame R15 chromanol as a safety-first comparator and SAR anchor, not as a ready topical efficacy candidate.

Keywords: chromanol, ADMET, hERG, skin permeability, Boltz-2, OpenMM, in silico, topical drug discovery

1. Research Question

This manuscript is intentionally narrower than the universal-scaffold paper. It asks whether the R15 chromanol fragment can be advanced as a safety-first scaffold and how that path should be separated from R16 topical chromanol optimization. Combining both paths would overstate the evidence: R15 has the cleaner predicted systemic-safety profile, while R16 analogs are better topical-window candidates but require separate safety and prior-art caveats.

2. Data Sources

file	role
pilot/cpu_meaningful/ r15_master_triage.csv	ADMET/xTB and topical composite ranking for 38 R15-derived candidates
pilot/cpu_meaningful/ r15_chromanol_cofold_14targets.csv	14-target Boltz-2 cofolding profile for R15_chromanol
pilot/md_r15_chromanol_top3_30ns/ summary.json	30 ns MD validation for the top-3 cofold targets
pilot/cpu_meaningful/ chromanol_pose_sanity_gate.csv	OpenMM/RDKit/PoseBusters raw-pose sanity gate
pilot/cpu_meaningful/ precompute_prior_art_gate.csv	technical prior-art pre-gate, not legal FTO opinion

3. Results

3.1 R15 is safety-first, not topical-first

label	SMILES	logP	QED	hERG	AMES	DILI	score	rank
R15 parent	<chem>OCC1COc2cc(O)ccc2C1</chem>	0.94	0.676	0.169	0.182	0.214	0.697	11
topical-score top-1	<chem>OCC1Cc2c(O)cc(O)cc2OC1C1COc2cc(O)cc2C1</chem>	1.97	0.665	0.616	0.101	0.408	0.752	1
topical-score top-2	<chem>CC(C)=CC1COc2cc(O)ccc2C1</chem>	2.91	0.713	0.575	0.259	0.202	0.748	2
topical-score top-3	<chem>CC(C)=CCC1COc2cc(O)ccc2C1</chem>	3.30	0.772	0.700	0.216	0.165	0.748	3

label	SMILES	logP	QED	hERG	AMES	DILI	score	rank
	O)ccc2C1							

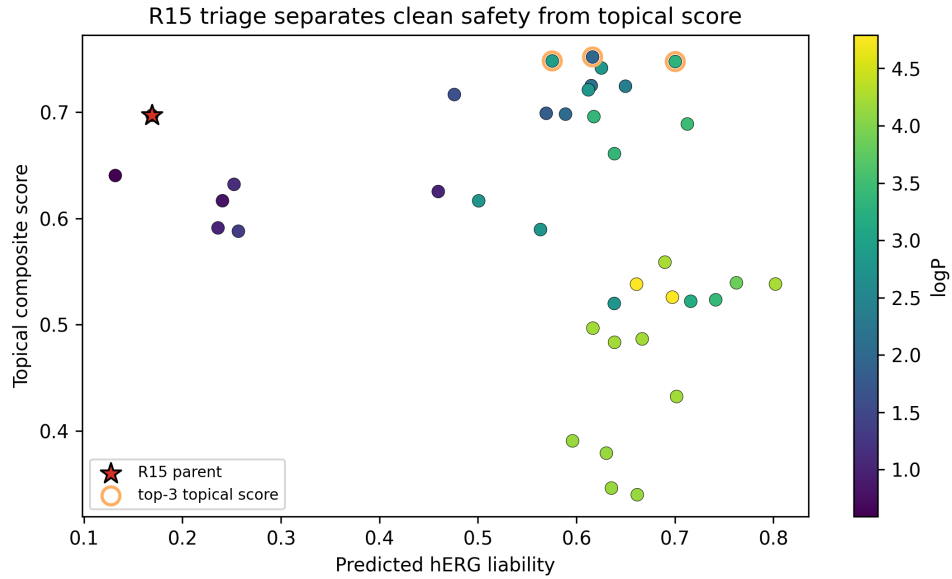


Figure 1. R15 triage scatter showing topical composite score versus predicted hERG. The R15 parent is highlighted separately from the top-3 topical-score candidates.

The parent fragment is useful because its predicted safety profile is cleaner than the topical-score leaders. It should not be described as the best topical lead. The correct narrative is a split path: R15 for safety-first fragment triage and R16 for topical optimization.

3.2 Cofolding identifies TGFB1, TYR, and DCT as top targets

rank	target	affinity probability	confidence score
1	TGFB1	0.594	0.709
2	TYR	0.536	0.892
3	DCT	0.502	0.887
4	PTGS2	0.494	0.908
5	SIRT1	0.463	0.515
6	TYRP1	0.460	0.896
7	AR	0.396	0.574
8	MMP1	0.393	0.719

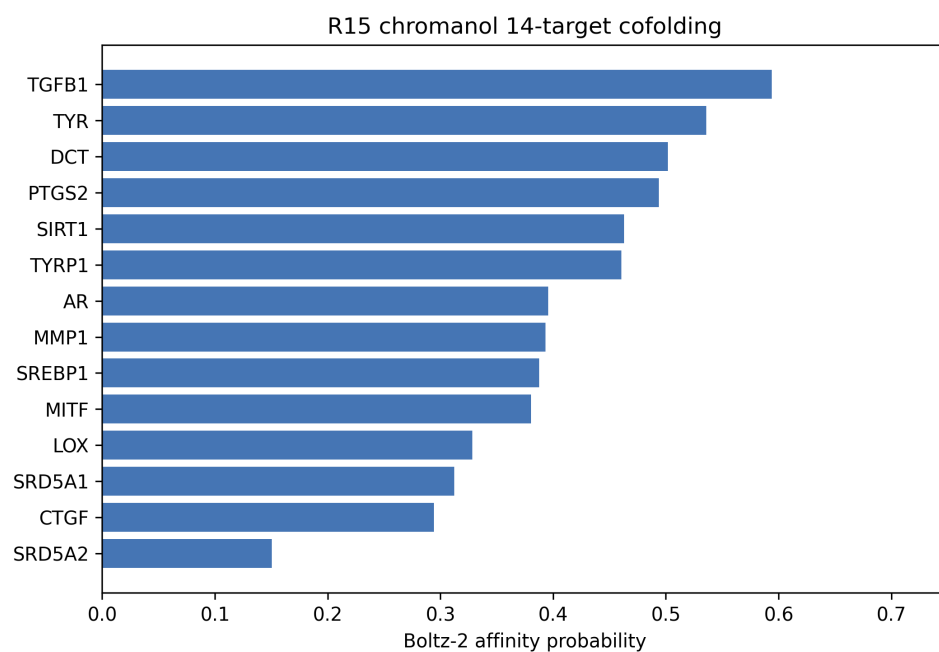


Figure 2. Boltz-2 cofold probabilities across 14 skin-relevant targets for the R15 chromanol parent.

3.3 Top-3 cofold targets persist over 30 ns MD

target	affinity probability	mean RMSD A	last-third RMSD A	max RMSD A
TGFB1	0.594	0.51	0.72	0.91
TYR	0.536	0.92	1.06	1.33
DCT	0.502	0.47	0.50	0.69

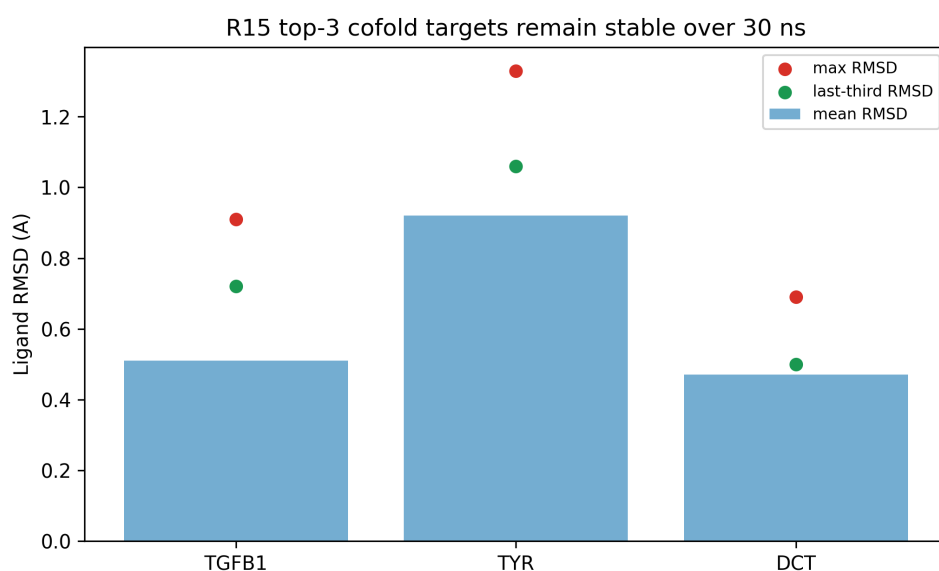


Figure 3. R15 top-3 target MD stability over 30 ns. Bars show mean RMSD and markers show maximum and last-third RMSD.

3.4 Raw-pose sanity gate is disclosed rather than hidden

Across the R15 parent 14-target panel, all poses loaded and were checkable. The pass/review/fail split was 9/5/0. Review status is not a rejection by itself; it means the raw cofold geometry has a caveat and should be interpreted together with minimization and MD persistence.

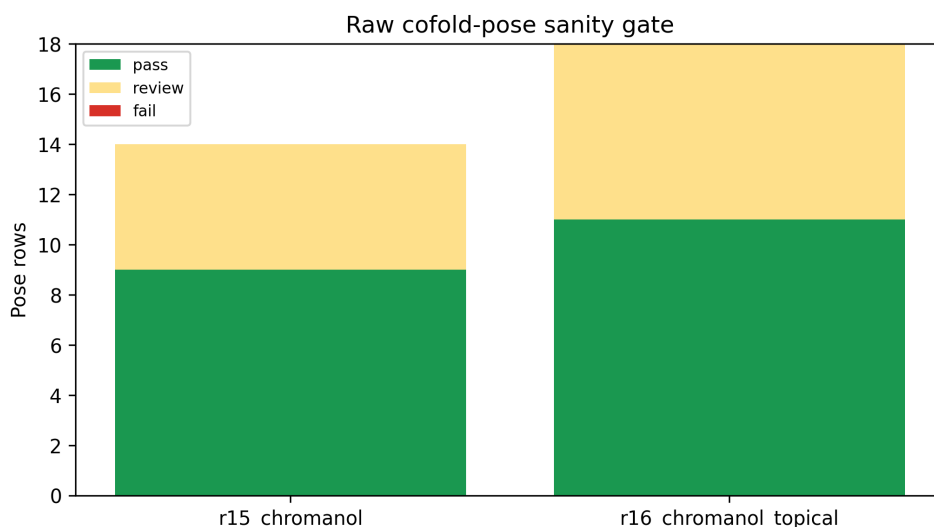


Figure 4. Raw cofold-pose sanity gate for R15 and R16 chromanol rows.

4. Prior-Art and Claim Discipline

The local precompute gate reports R15 parent PubChem exact status as hit and gate status as `hold_expensive_compute_until_prior_art_review`. This is a technical screen only. It does not establish freedom to operate or composition novelty. Any manuscript or commercial narrative must avoid clinical efficacy, confirmed target engagement, and FTO language until professional patent/Markush and wet-lab review are complete.

5. Limitations

All findings are in silico only. Boltz-2 cofolding is a prioritization model, not a binding assay. MD pose stability is not potency. ADMET-AI estimates cannot replace AMES, hERG patch-clamp, hepatotoxicity, irritation, sensitization, or permeation assays. Skin-window interpretation is heuristic and must be checked by formulation-dependent IVRT/IVPT and PBPK modeling.

6. Conclusion

R15 chromanol is best written as a safety-first fragment triage paper. Its strongest contribution is not a topical lead claim, but the disciplined separation of a clean predicted safety profile from the R16 topical optimization path.